Math is Finding Patterns — Homework

Lesson 1-5 • Standard 5.OA.B.3

Name: _____ **Period:** _____ **Date:** _____

Today's Goals

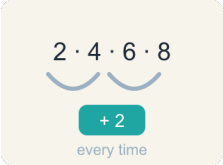
Content: I can find patterns and pattern rules, use them to make generalizations, and check my solutions with a table of values.

Language: I can describe a pattern using the words pattern rule, generalization, and 'each number is ___ more than the one before it.'

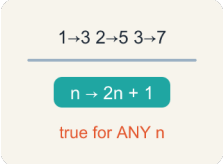
Key Words



pattern (patrón) — Something that repeats or changes in a predictable way.



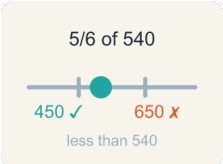
pattern rule (regla del patrón) — The instruction that tells how to get from one number in a pattern to the next.



generalization (generalización) — A statement that is true for a whole pattern, made by noticing repeated calculations.



table of values (tabla de valores) — A table that lists each step of a pattern so you can check that a solution is reasonable.



reasonableness (sensatez del resultado) — Whether an answer makes sense — checked as you work, with adjustments made when needed.

Practice

Show your work. Use the blank space under each problem.

1. Each penny has a value of $0.01. What is the value of 100 pennies?

A. $0.10
B. $1.00
C. $10.00
D. $100.00

2. Each quarter has a value of $0.25. Following the pattern, what is the value of 100 quarters?

A. $2.50
B. $25.00
C. $100.25
D. $250.00

3. Player A had 18 points before making 10 two-pointers in a row. How many points does Player A have after the streak?

A. 20
B. 28
C. 38
D. 48

4. Akela's jar totals $53.50. Why can the number of pennies only be a multiple of 5?

A. Because the total's cents end in 0, and only pennies make amounts that aren't multiples of 5 cents
B. Because pennies always come in rolls of 5
C. Because there are exactly 5 kinds of coins in the jar
D. Because $53.50 is divisible by 5

5. Which combination of coins totals exactly $53.50?

A. 100 quarters, 25 dimes, 15 nickels, 25 pennies
B. 200 quarters, 25 dimes, 15 nickels, 25 pennies
C. 200 quarters, 25 dimes, 15 nickels, 20 pennies
D. 200 quarters, 30 dimes, 15 nickels, 25 pennies

6. How does a table of values check that Player A really scored 38 points?

A. It rounds the answer to the nearest ten
B. It replaces the pattern rule so you don't need one
C. It proves the answer without doing any calculations
D. It lists the score after every shot, so you can see the pattern land exactly on 38

7. Use the multiple-of-5 pattern to find a DIFFERENT combination of coins that totals $53.50. Show that it works.

8. After the streaks, Player A has 38 and Player B has 36. Player B started 12 points behind but nearly caught up. What about the two pattern rules explains that?

9. Player A starts with 18 points and makes two-pointers. Complete the table.

Two-pointers made	Rule: 18 + 2 × shots	Total points
1	18 + 2 × 1	

Two-pointers made	Rule: 18 + 2 × shots	Total points
5	18 + 2 × 5	
10	18 + 2 × 10	